

Notebook: Credit Card Fraud - Cleaning, Feature Engineering, and EDA

Objectives:

- Clean the Dataset
- Engineer helpful features(amount transforms, time-based features)
- Explore patterns via visualizations(distributions, fraud rates, correlations)
- Summarize findings that will guide modeling choices

Findings Summary:

- Class Imbalance- Fraud rate is very low, metrics like PR AUC and recall- focussed thresholds in modeling.
- Amount : Distributions are highly skewed ,log transforms (Amount_log) help reveal tail behavior
- Temporal patterns: Hour and DaysSegment may show pockets of higher fraud rates, useful for simple risk signals
- Correlations: Some PCA components may show stronger association with the target, these can guide feature selections and modeling focus

This is the result I found with 4 models.

Based on the result, best performance model is KNN

	Model	Train Accuracy	Test Accuracy	Precision	Recall
0	KNN	0.9996	0.9994	0.9552	0.6737
1	Logistic Regression	0.9766	0.9753	0.0565	0.8737
2	SVM	0.9992	0.9985	0.5372	0.6842
3	Decision Tree	1.0000	0.9990	0.7126	0.6526

	F1 Score	ROC-AUC	PR-AUC	Fit Time
0	0.7901	0.8893	0.7566	0.2217
1	0.1061	0.9695	0.6735	3.7500
2	0.6019	0.9759	0.5743	1249.6944
3	0.6813	0.8261	0.4657	15.9090

